There are two basic ways of specifying to tolerance

1. Unilateral Tolerance
2. Bilateial Tolerance

UNILATERAL TOLERANCE

In this system, the tolerance is allowed to only one side of the basic size. Parts manufactured will fall close to the desired dimension but can vary in only one direction.

Example

Component Size	Basic size	Maximum limit	Minimum limit
$20^{+0.02}_{-0.00}$	20.00mm	20.02mm	20.00mm
$20^{+0.00}_{-0.02}$	20.00mm	20.00mm	19.98mm

BILATERAL TOLERANCE

In this system, the tolerance is allowed to both sides of the basic size. One limit will be above basic size and other limit below the basic size.

Example

Component Size	Basic size	Maximum limit	Minimum limit
$35^{+0.02}_{-0.02}$	35.00mm	35.02mm	34.98mm

8.6 FITS

The relation between two parts where one is inserted into the other with a certain degree of tightness or looseness is known as fit.

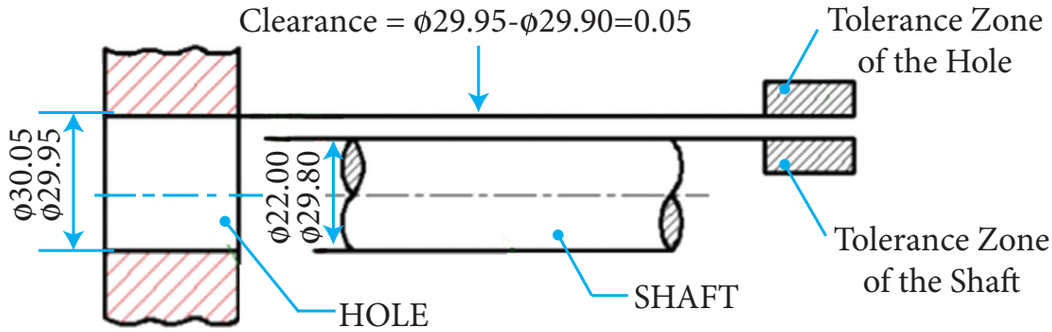
Types of Fits: Depending upon actual limits of the hole and the shaft, fit can be divided into three classes.

1. Clearance Fit
2. Interference Fit
3. Transition Fit

Clearance Fit: If a positive clearance exists between the hole and shaft, then it is called clearance fit. In this type of fit minimum clearance is always grater than zero. Such fits give loose joining and there will be some amount of freedom between the shaft and holes.

Example Bush bearings and channel bearings are fitted with clearance fit.

Interferance Fit: If a negative clearance exists between the hole and shaft, then it is called as "Interferance fit". In this type of fit, the shaft being larger than the hole. There are three grades of interferance fit namely,



Clearance Fit

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